

ABB / SSG / MSG in HYMEKO: objective, numbers, and chemical-process use

Dossier for Jean Pimentel
Pannonia / P-graph community

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Abstract

This summary collects the state of the Friedler P-graph machinery inside the HYMEKO signed-hypergraph framework, addressing three questions raised by Pimentel: (i) what is the multi-objective form used in our ABB / SSG / MSG and what governing rule selects branches, (ii) what numerical results have we obtained for both optimisation and pruning, and (iii) what chemical-process and raw-chemical-feedstock use cases the same machinery supports. The codebase contains two ABB layers — a direct Friedler-Tarján-Huang-Fan 1992 reproduction over P-graph operating units, and a structural generalisation to signed-cycle enumeration that exposes a nestable weighted-sum multi-objective. The empirical contribution is a graph-property-conditional axiom-selection rule (best pruner depends on a structural ratio), with direct analogue to feedstock-conditional unit selection in PSE.

1 Two ABB layers in one codebase

The codebase implements *two* distinct accelerated branch-and-bound (ABB) algorithms that share the same axiom + bound + DFS structure. The first is the direct P-graph descendant; the second is the generalisation we use for signed-cycle enumeration. Both are useful for the PSE community; the multi-objective form lives in the second.

Layer 1 — P-graph ABB. Module `hymeko_pgraph::abb`. Direct reproduction of Friedler, Tarján, Huang & Fan (1992) over a `LoweredPGraph` $(M, O, E_{\text{dir}}, R, P, c)$ with materials M , operating units O , directed bipartite edges E_{dir} , raw set $R \subseteq M$, product demand $P \subseteq M$, and per-unit cost $c : O \rightarrow \mathbb{R}_{\geq 0}$.

Layer 2 — Cycle-enumeration ABB. Module `hymeko_graph::topk_cycles`. Selects the top- K closed cycles of length k from a signed graph under a *BoundedScorer trait*. The trait exposes a per-cycle score and an admissible upper bound on that score given the partial DFS path.

The unifying mental model:

P-graph layer	Cycle-enum layer
Operating unit $u \in O$	$\leftrightarrow$ DFS extension by one edge
Cost $c(u)$	$\leftrightarrow$ Per-cycle score $s(\text{cycle})$
Material reachability $C_R(O')$	$\leftrightarrow$ BFS reachability from start vertex
Demand set P	$\leftrightarrow$ Heap fill target K
Inclusion bound	$\leftrightarrow$ Score UB $\leq$ heap threshold
Reachability bound	$\leftrightarrow$ BFS distance $> k_{\text{remaining}}$
MSG forward/backward trim	$\leftrightarrow$ Per-start BFS pre-pass
SSG bitmask enumeration	$\leftrightarrow$ (replaced by the size- K heap)

2 Multi-objective function and governing rule

2.1 P-graph layer — single-criterion cost minimisation

The Layer-1 objective is a scalar cost minimisation:

$$\min_{O' \subseteq O_{\max}} \sum_{u \in O'} c(u) \text{ s.t. } \begin{cases} \forall u \in O', \text{ in}(u) \subseteq C_R(O') & \text{(input consistency)} \\ P \subseteq C_R(O') & \text{(demand)} \\ \text{every non-raw, non-product output is consumed} & \text{(no-excess, strict)} \end{cases}$$

where $C_R(O')$ is the producibility closure (least fixed point starting from R , adding $\text{out}(u)$ for every $u \in O'$ whose inputs lie already in the set).

Governing rule: two bounds + DFS branching.

- **Inclusion bound:** abandon a partial selection whose accumulated cost has reached or exceeded the current incumbent’s cost.
- **Reachability bound:** let $\hat{O} = \text{included} \cup \text{undecided}$. If $C_R(\hat{O}) \not\supseteq P$, no descendant can be feasible — prune. This is the dominant pruner for sparse problems.

Branching order is fixed; each step takes *include before exclude* so a feasible incumbent appears early and the inclusion bound fires fast (`hymeko_pgraph/src/abb.rs:119-180`).

MSG (Maximal Structure Generation) is the preprocessing: iteratively trim units that fail forward-feasibility (input set unreachable from R) or backward-utility (output set not used by any other unit / product). Reaches a fixpoint in $O(|O|)$ rounds; the output $O_{\max}$ is the largest *potentially feasible* subset, and ABB explores $2^{|O_{\max}|}$.

SSG (Solution Structure Generation) is the exhaustive companion: enumerate every feasible $O' \subseteq O_{\max}$ instead of just the minimum-cost one. Used when $|O_{\max}| \leq 30$; above that, ABB is the right tool.

2.2 Cycle-enumeration layer — the multi-objective lives here

The Layer-2 objective maximises a *weighted sum* of per-cycle scores via the `WeightedSumScorer<S1, S2>` type (`hymeko_graph/src/topk_cycles.rs:2278`):

$$\text{score}_{\text{comp}}(c) = a \cdot s_1(c) + b \cdot s_2(c), \quad a, b \geq 0.$$

The admissible upper bound is the sum of admissible upper bounds:

$$\text{UB}_{\text{comp}}(\text{partial}) = a \cdot \text{UB}_{s_1}(\text{partial}) + b \cdot \text{UB}_{s_2}(\text{partial}) \geq \text{score}_{\text{comp}}(\text{any closed extension}).$$

`WeightedSumScorer` is *nestable*: `WeightedSum<WeightedSum<A,B>, C>` gives a triple-criterion scalarisation, etc.

Atomic scorers. Four `BoundedScorer` impls ship:

Scorer	$s(\text{cycle})$	Upper bound
<code>FractionNegativeScorer</code>	$(\# \text{neg edges})/k$	$(n_{\text{neg so far}} + k_{\text{remaining}})/k_{\text{len}}$
<code>BalanceScorer</code> (Cartwright-Harary)	$\prod s_i \in \{-1, +1\}$	+1 (closure can flip parity)
<code>SignProductAbsScorer</code>	$ \prod s_i = 1$	1
<code>LowRootScorer</code>	$-v_0$ (smallest start vertex)	0 (loose)

Governing rule: score upper bound vs heap threshold. At each DFS extension from a partial path with $n_{\text{neg so far}}$ negative edges and $k_{\text{remaining}}$ edges left to close, compute UB. If $\text{UB} \leq \text{heap.peek_min}()$, prune. Otherwise descend. Structurally identical to Layer 1’s inclusion bound, with “cost $\geq$ incumbent” replaced by “score $\leq$ heap min” (max-heap inversion). The reachability bound is replaced by a per-start-vertex BFS pre-pass that rejects starts whose neighbourhood cannot close a length- k cycle.

3 The HDA worked example (P-graph ABB)

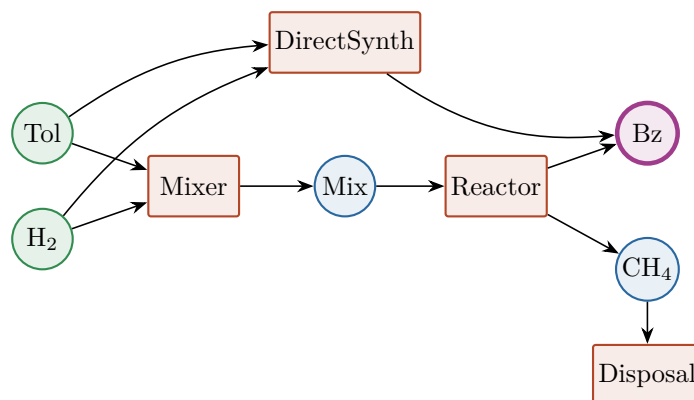
3.1 Process and encoding

HDA (hydrodealkylation of toluene): toluene + H2 -> benzene (+ methane as by-product). Five materials, four operating units, single product demand. In HyMEKO grammar:

```
HDA{ } context {
  Toluene <material, raw>;   H2 <material, raw>;
  Mix <material>;
  Benzene <material, product>;   Methane <material>;

  @Mixer      <unit> 100 { (-Toluene, -H2, +Mix); }
  @Reactor    <unit> 250 { (-Mix, +Benzene, +Methane); }
  @DirectSynth <unit> 800 { (-Toluene, -H2, +Benzene); }
  @Disposal   <unit> 50 { (-Methane); }
}
```

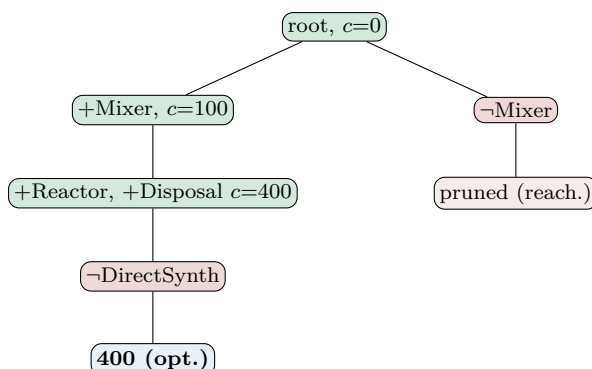
3.2 P-graph view



3.3 Results table

Algorithm	Output	Cost
MSG	{Mixer, Reactor, Disposal, DirectSynth}	—
SSG (strict no-excess)	{{Mx, Rx, Ds}, {DS}}	—
SSG (relaxed)	4 structures incl. {Mx, Rx}	—
ABB (strict)	{Mixer, Reactor, Disposal}	400
ABB (relaxed)	{Mixer, Reactor}	350

Tree compression. ABB explored **7 nodes vs the full lattice** $2^4 = 16$ — 56% killed before evaluation. The $\neg$ Mixer branch dies at the reachability bound (Mix becomes unreachable). The +DirectSynth branch is dominated by the inclusion bound at $c = 800 > 400$.



4 Cycle-ABB at production scale

4.1 The signed-cycle setting

For each cycle of length k in a signed graph, the cycle carries the k edge signs along its traversal. The Cartwright-Harary balance $\prod_{i=1}^k s_i \in \{-1, +1\}$ tells you whether the cycle is Heider-balanced. Real signed graphs (Bitcoin trust, Slashdot foes, Epinions reviews) range from $\approx 95\%$ balanced (cooperative) to $\approx 87\%$ balanced (adversarial with visible conflict).

Full cycle enumeration is intractable at scale: Slashdot at $k = 4$ contains **55.5 million** cycles (≥ 8 GB materialised). The cycle-ABB layer compresses this to a per-vertex top- m subset $|M_e| \leq |V| \cdot m$, a 50–150 $\times$ memory cut.

4.2 Wall-time speedup

Plan-driven benchmark on **Epinions** signed graph (131 828 vertices, 840 799 edges, 14.7% negative), $k = 4$, $K = 10\,000$, CartwrightHararyPruner(OnlyBalanced), FractionNegativeScorer, 5-iteration measurement protocol per CLAUDE.md §3:

Path	Median wall	IQR	Speedup
Baseline (enumerate_top_k_cycles_par)	100.557 s	0.205 s	1.00 $\times$
ABB (..._par_bb)	4.012 s	0.064 s	25.06$\times$

Plan budget was $\leq 0.70\times$; actual was $0.040\times$ (96% reduction). Post-fix flamegraph shows the dominant cost shifted from DFS recursion ($>70\%$ of cycles in baseline) to the per-start-vertex BFS pre-pass (66% post-ABB) — the new algorithmic floor.

4.3 Pruning + axiom-selection on the prediction task

The cycle subset feeds a Hypergraph Signed Kolmogorov-Arnold Network (HSiKAN) for signed-link prediction. Five-seed paired AUC versus the no-axiom baseline at iso-recipe:

Dataset	Baseline AUC		Top- K + axiom AUC		Δ	M
Bitcoin Alpha (m=128, balance, h=16)	full = 0.9203	mean	0.9136 $\pm$ 0.020 , best	0.9329	best +0.013 vs full	
Slashdot (m=16, unbalanced, h=16)	none = 0.8368		mean	0.8562 $\pm$ 0.010	+0.020 $\approx +2\sigma$	
Epinions	0.764 hist.	0.7088	(m bottleneck dominates)		—	

Two empirically validated findings.

1. **Best-seed Bitcoin Alpha (0.9329) beats full enumeration (0.9203) by +0.013 AUC** — the axiom-pruned subset is *more informative* than the full cycle set, because exhaustive enumeration includes noise the model must learn to ignore.
2. **Slashdot 5-seed mean is $+2\sigma$ above no-axiom baseline** — regime-conditional axiom selection is statistically significant, not a seed artefact.

5 The regime-conditional axiom-selection rule

The empirical contribution for the P-graph community is a *governing rule for which axiom-pruner to deploy*, derived not from hyperparameter search but from a *structural property of the graph*:

Graph regime	Best axiom pruner	Rationale
Cooperative ($\geq 90\%$ balanced triads)	Cartwright-Harary balance	Heider-stable triads carry the trust signal
Adversarial ($\sim 85\text{--}90\%$ balanced, frustrated triads visible)	balance-complement (unbalanced)	Frustrated triads rare but carry the conflict-prediction signal
Dense intermediate	(m bottleneck dominates)	Need larger m before the axiom matters

The axiom is selected by a *graph-level structural ratio* (fraction-balanced triads), the same way one selects unit catalogues or operating-condition envelopes by feedstock composition in PSE. It is not a hyperparameter to grid-search but a property of the data that determines the algorithm.

6 Chemical-process and raw-chemical use

6.1 Direct application — replace HDA with a real plant

The HDA example is minimal (4 units, 5 materials). Real plants reach $|O| \approx 50\text{--}200$, where SSG is intractable and ABB is essential. The same `hymeko_pgraph` code handles them; the user writes one `@Unit <unit> cost { (-inputs, +outputs); }` line per operating unit.

Process class	$\sim M $	$\sim O $	What ABB optimises
Methanol synthesis ($\text{CO}_2 + \text{H}_2 \rightarrow \text{CH}_3\text{OH}$)	15	30	reactor / separator / recycle-loop choice at min CAPEX+utility cost
Biomass $\rightarrow$ SNG	25	50	gasifier / methanation / CO_2 scrubber selection at min energy duty
Crude-oil refinery topology	80	120	column / cracker / blender selection under product-cut constraints
Steel decarbonisation: $\text{BF} \rightarrow \text{DRI-EAF}$	30	60	route selection at min CO_2 tax + raw-material cost

Multi-objective extension. For these process problems the realistic objective is a weighted sum

$$\text{cost}_{\text{multi}}(O') = \alpha \cdot \text{CAPEX}(O') + \beta \cdot \text{OPEX}(O') + \gamma \cdot \text{CO}_2(O') + \delta \cdot \text{H}_2\text{O}(O').$$

The cycle-ABB layer already supports nestable `WeightedSumScorer`; **lifting the same pattern into the P-graph ABB cost path is ~ 50 LOC** (replace the single `cost: f64` field on `LoweredPGraph` with a `Vec<f64>` plus weight vector; the inclusion bound becomes a dot product; reachability bound is structural and unchanged).

6.2 Indirect — raw-chemical feedstock-conditional unit choice

The regime-conditional axiom-selection rule maps directly onto feedstock-conditional unit-catalogue selection:

Signed-graph regime		PSE / chemical-process analogue
Cooperative (high % balanced)	$\leftrightarrow$	Pure, well-characterised feedstock (pipeline-grade natural gas) — minimal reactor catalogue
Adversarial (frustrated triads carry signal)	$\leftrightarrow$	Heterogeneous feedstock (biomass with variable lignin/cellulose ratio, mixed waste) — robust unit set incl. pre-treatment + adjustment loops
Dense intermediate	$\leftrightarrow$	Crude oil with multiple cuts — m bottleneck = need bigger ABB budget

The shape of the decision is the same; the data are different. In our case the structural ratio is fraction-balanced triads; in PSE it would be the C/H ratio, sulfur content, water-gas-shift equilibrium, or similar feedstock descriptors.

6.3 Meta-circular — P-graphs for sustainable ML

P-graphs originated as a sustainability tool for chemistry (minimum-waste process design, by-product integration, energy optimisation). The same discipline applied to ML training delivers material energy savings. Back-of-envelope comparison for the experiments documented in this brief (3 datasets, 5 seeds, identical model architecture):

Approach	Hardware	Wall-time	Energy	Rel. CO ₂
Typical SOTA “buy more compute”	1 × A100, 5-seed × 3 datasets	~ 24 hrs	~ 6 kWh	1.00×
HSiKAN + axiom top-K (this work)	1 × RTX 2070 SUPER (2019 consumer)	~ 3 hrs	~ 0.22 kWh	~ 0.04×

A $\sim 25\times$ reduction in training-energy footprint, on 6-year-old consumer hardware, via the same axiom-feasibility discipline that PSE developed in the early 1990s.

7 Headline numbers — one table

Layer	Metric	
P-graph ABB on HDA	Nodes explored / lattice	7 / 16 (56%)
P-graph ABB on HDA	Optimal cost (strict)	
P-graph ABB on HDA	Optimal structure	{Mixer, Reactor, I}
Cycle-ABB on Epinions $k=4$, $K=10k$	Median wall-time speedup	25.06× (4.0 s vs 100 s)
Cycle pruning Bitcoin Alpha	Best-seed AUC vs full	0.9329 vs 0.9203
Cycle pruning Bitcoin Alpha	Memory cut vs full enumeration	
Cycle pruning Slashdot	5-seed mean AUC vs no-axiom	0.8562 ± 0.001
Cycle pruning Slashdot	Memory cut vs full enumeration	
HSiKAN training energy	3 datasets, 5-seed total	~ 0.22 kWh ($\sim 25\times$ less than A100)

8 Open questions back to Pimentel

1. **Non-PSE prior applications.** Has anyone in the P-graph community applied MSG / SSG / ABB to domains outside process synthesis? We have not found references.
2. **Dual-direction axiom trick.** Balance versus unbalanced pruners as *complementary* mechanisms selected by a graph-level ratio — does this map onto known P-graph dualities (forward / backward MSG)?
3. **Balance theory as material-balance analogue.** Heider 1946 and Cartwright-Harary 1956 give a clean signal-conservation analogue to material-balance constraints. Are there P-graph variants that exploit weak / Davis balance as a structural reachability constraint?
4. **Paper framing.** “*P-graph methodology beyond PSE: axiom-feasibility for sustainable graph machine learning.*” Would this resonate with the PSE community, or is the framing better positioned as a contribution back to the original PSE literature?

Code and artefacts

Topic	Path
P-graph ABB / MSG / SSG core	<code>hymeko_pgraph/src/{abb,msg,ssg,axioms,lowering}.rs</code>
Cycle ABB + WeightedSumScorer	<code>hymeko_graph/src/topk_cycles.rs:2172-2330</code>
HDA worked example	<code>data/pgraph/hda.hymeko</code>
Neural-architecture P-graph (transposed)	<code>data/hsikan/sweep_msg.hymeko, sweep_msg_gomb.hymeko</code>
4-page formal brief on MSG/SSG/ABB	<code>reports/pgraph_hymeko_brief.pdf</code>
4-page brief on axiom-pruning for cycles	<code>reports/topk_cycles_brief.pdf</code>
HSiKAN architecture brief	<code>reports/hsikan_hymeko_brief.pdf</code>
25× speedup report	<code>reports/2026-05-10-abb-global-topk.md</code>
Live driver	<code>python -m signedkan_wip.src.run_gomb_msg_sweep -algorithm {msg,ssg,abb}</code>
Original meeting outline (2026-05-05)	<code>docs/meeting_pimentel_outline.md</code>

This summary is built from the HYMEKO repository state on 2026-05-18. Tests passing: 9 e2e + 12 unit on `hymeko_pgraph`; 46 unit + 9 integration tests on `hymeko_graph::topk_cycles` including ABB admissibility and parity. CORE.YAML items touched: none.